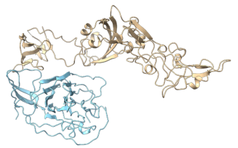
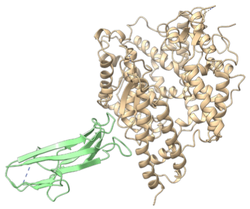
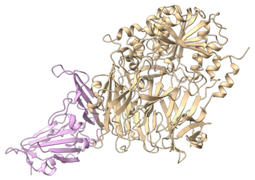
## 1. Target selection & structure retrieval



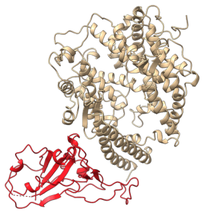
HCoV-HKU1
(PDB: 8Y7Y)



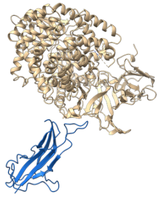
HCoV-NL63
(PDB: 3KBH)



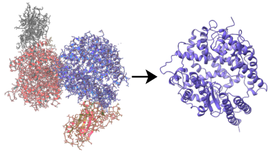
MERS-CoV
(PDB: 4KR0)



SARS-CoV-1
(PDB: 2AJF)



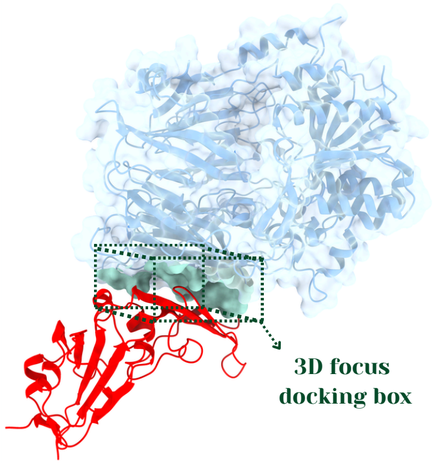
HCoV-229E
(PDB: 6ATK)



- Remove non-receptor chains
- Remove waters
- Remove ligands

## 2. Receptor Interface hotspot mapping

Dipeptidyl peptidase 4



3D focus docking box

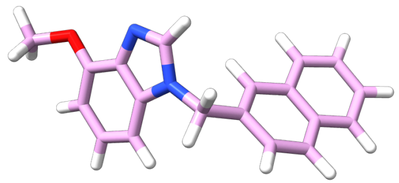
protein receptor
(MERS-CoV spike RBD)

Residues within 4.0 Å of the receptor = hotspot residues

Defines the receptor–interface docking region

## 3. GNINA molecular docking

CKP-25 compound



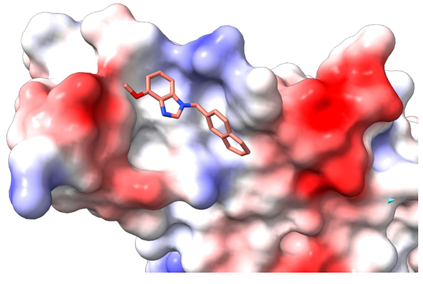


Ligand preparation & minimization

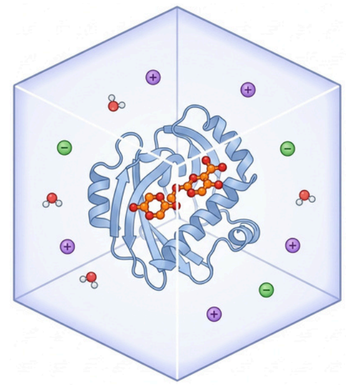


- GNINA docking
- Exhaustiveness = 8

Best docked pose of CKP-25



## 4. (x3) 100 ns explicit solvent MD



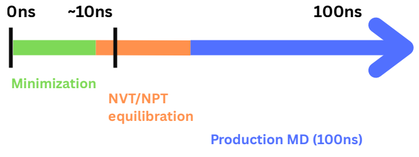
Na⁺ Cl⁻ Water

Solvated CKP-25-spike complex

Energy minimization

NVT/NPT equilibration

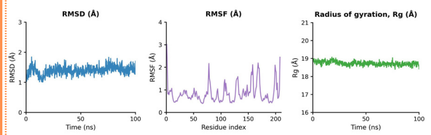
(x3) 100-ns production run




## 5. Results & Analysis

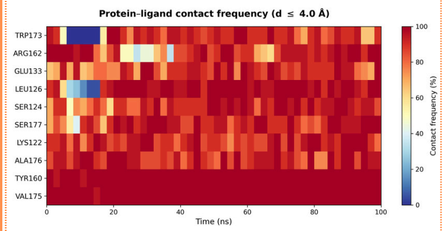
*MM-GBSA analysis performed on the stable 50–100 ns window*

### System Stability

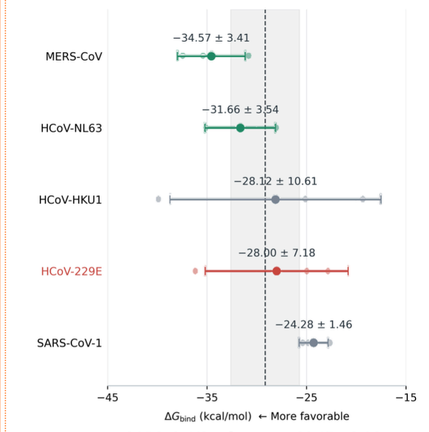



*System remains stable during 50 - 100 ns.*

### Protein - ligand contact frequency (d ≤ 4 Å)




### MM-GBSA binding estimates




Interface retained in 4/5 targets

HCoV-229E relocated in 2/3 MD runs